Table 1.9 Aramid fibres

Fibre	Properties	Uses
Kevlar®	Lightweight, flexible and extremely durable aramid; excellent resistance to heat, corrosion and damage from chemicals and high tensile strength-to-weight ratio	Used in protective clothing such as police body armour, where the fibre is woven in a lattice that provides protection against knife attack
Nomex®	Extremely strong synthetic fabric that can withstand exposure to the most extreme conditions	Primarily used where resistance to heat and flames is essential, for example firefighters' uniforms

Sports textiles

- **Micro-encapsulation** is a process of applying microscopic capsules to fibres and fabrics.
- The capsules can contain vitamins, therapeutic oils, moisturisers, antiseptics and antibacterial chemicals that are released through friction.
- Rhovyl® is a non-flammable, synthetic fibre that is crease resistant, has good thermal and acoustic properties, is antibacterial and comfortable to wear.
- The construction of the fibre gives fabrics the ability to wick away moisture, such as perspiration, through the fabric. It also dries quickly, meaning it does not retain odours, making it ideal for socks and sportswear.

Micro-encapsulation: tiny microscopic droplets containing various substances applied to fibres, yarns and materials including paper and card.

Now test yourself

TESTED

1 Describe what gives plywood its strength. (2)
2 Explain why carbon fibre reinforced polymer (CFRP) is a suitable material for a racing bike. (3)
3 Explain how micro-encapsulation is beneficial to a patient when used in medical dressings. (2)
4 State what is meant by 'smart glass'. (1)
5 Describe **two** reasons why the aramid fibre Nomex® is used in protective clothing. (2)

1.5 Mechanical devices

1.5.1 Types of movement

REVISED

The four main types of movement are:

- **linear**: movement in a straight line
- **reciprocation**: movement that goes back and forth in a straight line
- **rotary**: movement that turns in a circle
- **oscillation**: a swinging movement from side to side.

1.5.2 Classification of levers

REVISED

- **Levers** consist of a rod pivoted on a **fulcrum**.
- In a class 1 lever, the effort and the load are on opposite sides of the fulcrum. An example is a see-saw.
- In a class 2 lever, the load and the effort are on the same side of the fulcrum, with the load nearest the fulcrum. The effort required is less than the load. An example is a wheelbarrow.
- In a class 3 lever, the load and effort are on the same side of the fulcrum, but this time the effort is nearest to it. The effort required is greater than the load, but the load can be moved faster. An example is a pair of tweezers.

Lever: a rod pivoted on a fulcrum – the class of lever (1, 2 or 3) affects how much effort is needed to move the load.

Fulcrum: the point, or pivot, on which a lever sits.

Exam tip

Make sure that you show all of your working when performing calculations related to mechanical systems, including writing down the formula.

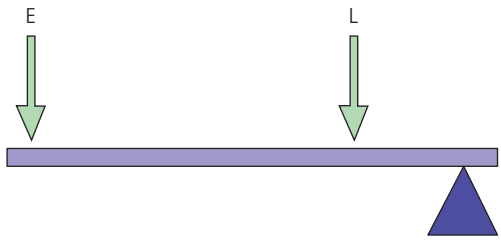


Figure 1.11 A class 2 lever

$$\text{Mechanical advantage of a lever} = \frac{\text{Load } (F_b)}{\text{Effort } (F_a)}$$

$$\text{Velocity ratio of a lever} = \frac{\text{Distance moved by effort}}{\text{Distance moved by load}}$$

$$\text{Efficiency of a lever} = \frac{\text{Work done on the load}}{\text{Work done by the effort}} \times 100$$

1.5.3 Linkages

REVISED

- **Linkages** are used to change the size of a force, the direction of motion and/or the type of motion.
- In a **reverse motion linkage**, the bottom link moves to the left as the top link moves to the right. If the distance between the fixed pivot and the moving pivots is equal, then the output force is the same as the input force. The output force can be increased or decreased by changing the position of the fixed pivot.
- In a **bell crank** the output movement is at 90 degrees to the input movement. The output force is greater than the input force when the fixed pivot is closer to the output lever.

Linkage: levers connected together via fixed and moving pivots; they are used to change the size of a force and/or the direction of motion.

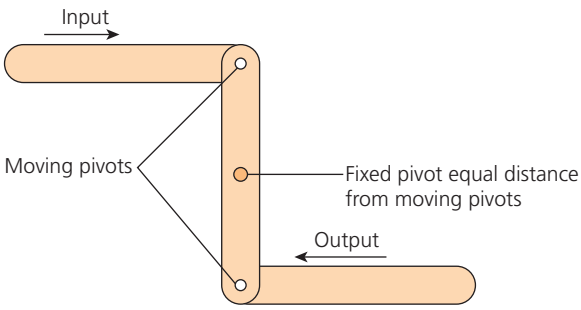


Figure 1.12 A reverse motion linkage

1.5.4 Cams

REVISED

- **Cams and followers** turn rotary motion into reciprocating motion. The follower moves up and down as the cam rotates.
- With a **pear-shaped cam**, the follower remains stationary (dwells) for half of the cycle. It is then pushed up as the point of the cam approaches. Finally, as the point passes, the follower falls and then dwells again.
- **Eccentric cams** are circular but have an off-centre rotating shaft. The rise and fall produced is symmetrical.
- With a **snail/drop cam** the follower dwells for the first 120 degrees or so of the cycle. It then rises slowly before dropping suddenly when it reaches the peak of the snail shape.

Cam and follower: a mechanism that converts rotary motion into reciprocating motion.

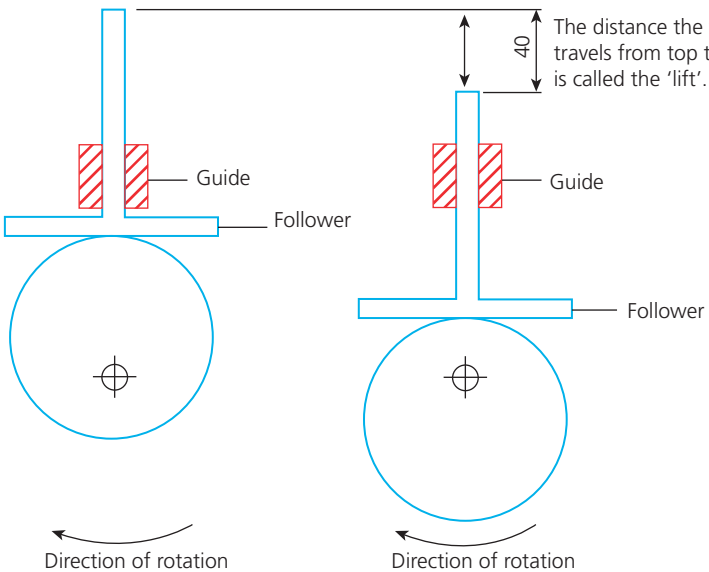


Figure 1.13 An eccentric cam mechanism

1.5.5 Followers

REVISED

- **Knife followers** have a sharp point that makes contact with the cam. They often wear quickly, leading to the mechanism not working correctly.
- **Roller followers** replace the sliding motion between the cam and follower with a rolling motion. This reduces wear.
- **Flat followers** are used where space is limited.

1.5.6 Pulleys and belts

REVISED

- V-belt pulley systems transmit rotary motion and torque. A larger pulley driving a smaller pulley results in increased speed but decreased torque. A smaller pulley driving a larger pulley achieves the opposite.
- The speed of a pulley is measured in revolutions per minute (rpm).

Pulley and belt: A system that uses pulley wheels and a belt to transmit rotary motion and torque.

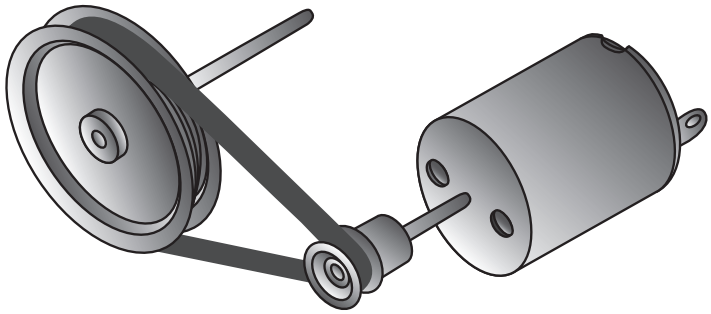


Figure 1.14 An electric motor driving a simple pulley system

Typical mistake

It is easy to mix up the driven pulley and the driver pulley when calculating velocity ratios. Ensure you know which one is which.

$$\text{Velocity ratio (VR) of a pulley system} = \frac{\text{Diameter of the driven pulley}}{\text{Diameter of the driver pulley}}$$

$$\text{Speed of the driven pulley (rpm)} = \frac{\text{rpm of the driver pulley}}{\text{Velocity ratio}}$$

1.5.7 Cranks and sliders

REVISED

- **Crank and slider** mechanisms convert rotary motion to reciprocating motion, and vice versa. To create reciprocating motion, the crank rotates and the connecting rod pushes the slider backwards and forwards.
- Piston engines in cars are an example of the reverse.

Crank and slider: a mechanism that converts rotary motion to reciprocating motion and vice versa.

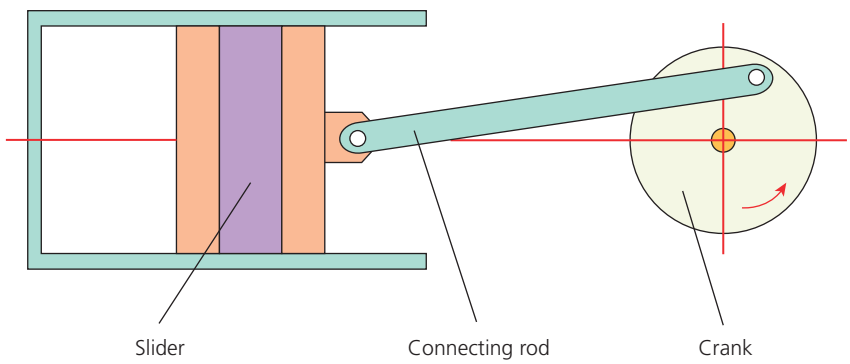


Figure 1.15 A crank and slider mechanism

1.5.8 Gear types

REVISED

- In a simple two **gear train**, a larger gear driving a smaller gear results in increased speed but decreased torque. A smaller gear driving a larger gear achieves the opposite.

$$\text{rpm of the driven gear} = \frac{\text{rpm of the driver gear}}{\text{Gear ratio}}$$

Gear train: a collection of spur gears that transmit rotary motion and torque.

- **Compound gear trains** are created when one or more of the shafts holds two gears. They are used when large increases in speed or torque are needed and when space is limited.
- An **idler gear** is placed between two gears to ensure that they have the same direction of motion.
- **Bevel gears** are used to change the direction of a shaft's rotation, usually through 90 degrees.
- **Rack and pinions** change rotary motion to linear motion. The teeth on the circular pinion gear mesh with those on the rack. As the pinion gear turns, the rack moves in a straight line.

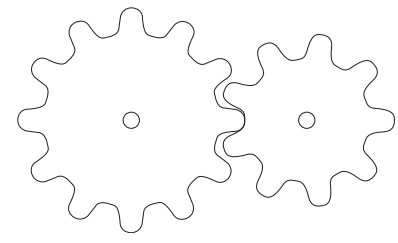


Figure 1.16 A simple gear train